Recovering a time-dependent control field from qubit expectation values: physics-informed neural networks versus classical inversion

Gubio G. de Lima[1, *]

[1] *Physics Department, Federal University of São Carlos, 13565-905, São Carlos, SP, Brazil*

(Dated: November 6, 2026)

The control field that actually drives a qubit differs from the programmed one, and recovering it from measured data is an inverse problem. We recover an unknown time-dependent field $f(t)$ of a driven qubit, $H = \frac{1}{2}\sigma_z + \frac{1}{2}f(t)\sigma_x$, from the time traces of $\langle\sigma_x\rangle$, $\langle\sigma_y\rangle$ and $\langle\sigma_z\rangle$, and compare physics-informed neural networks (PINNs) with classical estimators under a single protocol: identical data, initial state $|0\rangle$, no access to the true field, five random seeds, and two metrics (field error and re-simulated trajectory error). Four representations of the field are studied—a neural network, a Fourier series, B-splines and the MixFunn architecture—together with a cosine series. PINNs trained with Adam alone, as in common practice, converge to an over-smoothed field with $\simeq 25\%$ error, whatever the representation, while the re-simulated trajectory already agrees with the data. A Levenberg–Marquardt (Gauss–Newton) stage after Adam lowers the error by two orders of magnitude, to $\simeq 0.2\%$: the optimizer, not the representation, is the bottleneck. With complete and noiseless data, classical linear least squares and shooting with a cubic B-spline basis remain about ten times more accurate and thousands of times faster; with noise, all good methods are limited by the information in the data. We also show that an initial state aligned with the control axis makes the field unidentifiable, and we identify analogous blind states and symmetries for a two-qubit exchange coupling and for partial measurements, which define where PINNs can be expected to help. **[TODO:** co-authors and affiliations; update numbers when the benchmark is complete**]**

I. INTRODUCTION

Quantum gates are implemented by time-dependent control fields, and the field that reaches the qubit is distorted by the transfer function of the control lines, by crosstalk and by drifts of the device parameters [1]. Characterizing the dynamics from measured data—Hamiltonian learning—is therefore a central task in the calibration of quantum hardware [2, 3]. When the unknown is a whole function of time, rather than a few constant parameters, the task becomes an inverse problem for a differential equation: find $f(t)$ such that the solution of the equation of motion reproduces the measured expectation values.

Physics-informed neural networks (PINNs) [4, 5] solve such inverse problems by representing the solution with a neural network and penalizing the residual of the governing equation together with the misfit to the data. In quantum physics, PINNs have been used to design control fields [6], to learn general two-qubit Hamiltonians with a freezing mechanism [7], to identify Hamiltonian parameters and decay rates of open systems [8], and to approach the Heisenberg limit in Hamiltonian learning [9]. In these works the unknowns are constant parameters or the field is a design variable; recovering an *unknown time-dependent* field from data, and assessing how a PINN compares with the estimators that a physicist would use first, has received less attention.

Two facts make such a comparison necessary. First, PINN losses are ill-conditioned, and first-order optimizers such as Adam often stop far from the minimum [10, 11]; quasi-Newton and Gauss–Newton methods have recently been shown to improve PINN accuracy by orders of magnitude [12–16]. A disappointing PINN result may therefore reflect the optimizer rather than the method. Second, when all components of the Bloch vector are measured, the field can also be estimated by classical means: pointwise inversion, linear least squares on a basis, or shooting [17].

Here we compare, under a single protocol, PINNs with four representations of the field—a neural network, a Fourier series, B-splines [18] and MixFunn [19]—and a cosine series, trained with several optimizers, against classical estimators. Our main findings are: (i) an initial state aligned with the control axis makes the field unidentifiable at early times, which we avoid by preparing $|0\rangle$; (ii) Adam-only PINNs converge to an over-smoothed field for every representation, while a Levenberg–Marquardt (LM) stage lowers the field error by two orders of magnitude; (iii) with complete noiseless data, classical estimators with a good basis remain more accurate and much cheaper, and with noise all good methods are limited by the data. We close by identifying the blind states and symmetries of two harder settings—a two-qubit exchange coupling and partial measurements—where PINNs can be expected to be useful.

Section II states the problem and analyzes its identifiability. Section III describes the representations of the field and Sec. IV the estimators and the protocol. Results for the three test fields are presented in Sec. V, the outlook in Sec. VI, and conclusions in Sec. VII. Implementation details and all numerical values are given in the Supplemental Material (SM).

* gubio.gomes@estudante.ufscar.br

II. PROBLEM AND IDENTIFIABILITY

A. Driven qubit

We consider a qubit driven by an unknown field $f(t)$,

$$H(t) = \tfrac{1}{2}\,\sigma_z + \tfrac{1}{2}\,f(t)\,\sigma_x, \tag{1}$$

in units where $\hbar = 1$ and the qubit splitting is one, so that time is measured in units of the inverse splitting. The Bloch vector $\boldsymbol{r} = (X, Y, Z) = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$ obeys the Bloch equations [20, 21] $\dot{\boldsymbol{r}} = \boldsymbol{\Omega}(t) \times \boldsymbol{r}$ with $\boldsymbol{\Omega} = (f, 0, 1)$, i.e.,

$$\dot{X} = -Y, \qquad \dot{Y} = X - fZ, \qquad \dot{Z} = fY. \tag{2}$$

The data are $N = 100$ equally spaced samples $\boldsymbol{r}_i = \boldsymbol{r}(t_i) + \boldsymbol{\eta}_i$ on $[0, T]$, with Gaussian noise $\boldsymbol{\eta}_i$ of standard deviation σ. We use three test fields [Fig. 1(a)–(c)]: field 1, $f = \cos t$ with $T = 2\pi$, a single harmonic; field 2, $f = 1.5\sin 2t + 0.8\cos(3.5t + 0.5) + 0.4\sin 7t$ with $T = 2.52\pi$, which is not periodic on $[0, T]$; and field 3, the same kind of three-tone signal multiplied by the envelope $\sin^2(\pi t/T)$, a smooth pulse that starts and ends at zero.

B. Identifiability and choice of the initial state

The field enters Eq. (2) only through $fZ = X - \dot{Y}$ and $fY = \dot{Z}$. At each instant, the least-squares solution of these two equations is

$$f(t) = \frac{Z\,(X - \dot{Y}) + Y\,\dot{Z}}{Y^2 + Z^2}, \tag{3}$$

and the sensitivity of the dynamics to f is $\|\partial\dot{\boldsymbol{r}}/\partial f\|^2 = Y^2 + Z^2 = 1 - X^2$ for a pure state. When the Bloch vector points along the control axis σ_x, the field has no effect and cannot be identified. The state $|+\rangle$, $\boldsymbol{r} = (1, 0, 0)$, is the worst possible choice: at $t = 0$ the data carry no information about f. With $|0\rangle$, instead, $\dot{Y}(0) = -f(0)$ and $Y^2 + Z^2 \geq 0.13$ over the whole interval for the three test fields [Fig. 1(g)–(i)]. With the same measurement budget, preparing $|0\rangle$ instead of $|+\rangle$ halves the error of the classical estimators and strongly reduces its spread over noise realizations (Table S3 of the SM). We use $|0\rangle$ throughout.

Two further properties limit what can be recovered. The map $f \mapsto \boldsymbol{r}$ is smoothing: a perturbation $a\sin\omega t$ of the field changes the trajectory by an amount that decays as $1/\omega$ (Fig. S1), so the high-frequency content of f is weakly encoded in the data and is the first casualty of noise or of an inaccurate fit. And with partial data the problem may have exact symmetries: because $\sigma_z H(f)\sigma_z = H(-f)$ and $\sigma_z|0\rangle = |0\rangle$, the fields f and $-f$ produce identical $\langle\sigma_z\rangle(t)$ (Sec. VI B).

III. REPRESENTATIONS OF THE FIELD

All estimators below search for f in a family $\{f_{\boldsymbol{\theta}}\}$. The choice of family fixes which fields can be represented, how smooth the estimate is, and whether the problem is linear in the parameters. We study four representations used in the PINN literature, plus a cosine series that removes a known defect of the Fourier series.

a. Neural network. The field is an output of a fully connected network, $f_{\boldsymbol{\theta}}(t) = \mathcal{N}_{\boldsymbol{\theta}}(t)$. By the universal approximation theorem [22, 23], such networks can represent any continuous field on $[0, T]$, and the representation adapts its resolution to the data without a predefined grid. Gradient-trained networks, however, learn low frequencies first (spectral bias) [24], which is precisely the regime where the data are least sensitive (Sec. II B); input embeddings such as Fourier features mitigate this bias [25]. As in the original inverse PINN [4], we let the field be an extra output of the network that represents the state, so that state and field share their hidden layers.

b. Fourier and cosine series. A truncated Fourier series, $f(t) = a_0 + \sum_{m=1}^{M}[a_m\cos(2\pi mt/T) + b_m\sin(2\pi mt/T)]$, converges spectrally fast for smooth *periodic* functions [26]. If $f(0) \neq f(T)$, its periodic extension is discontinuous and the series suffers from the Gibbs phenomenon: convergence becomes algebraically slow and the error concentrates at the edges [26]. The cosine series $f(t) = \sum_{m=0}^{M} c_m\cos(m\pi t/T)$ represents the even extension of f to period $2T$, which is continuous for any f, and avoids this problem. The maximum frequency of the two series is fixed by M ($2\pi M/T$ and $\pi M/T$, respectively).

c. B-splines. A spline of degree k is a piecewise polynomial with $k-1$ continuous derivatives at the knots; it can be written as $f(t) = \sum_{j=1}^{n} c_j B_{j,k}(t)$ in the B-spline basis, computed by the Cox–de Boor recursion [18]. B-splines have local support ($k + 1$ knot intervals), form a partition of unity, and approximate smooth functions with error $O(h^{k+1})$ for a knot spacing h [18]. Smoothness can be controlled by penalizing differences of neighbouring coefficients (P-splines) [27]. B-splines are also the building block of Kolmogorov–Arnold networks [28], recently trained as physics-informed models [13]. We use cubic splines ($k = 3$) with open uniform knots.

d. MixFunn. MixFunn [19] is a network whose neurons apply a dictionary of functions (sin, cos, $e^{-0.01|x|}$ and the identity) to linear projections of their input and mix them with learnable weights; second-order neurons add products of these functions. The resulting models are compact and can be converted into closed-form expressions, which makes them attractive when the field is expected to be a combination of a few elementary functions.

e. Linear and nonlinear representations. The Fourier, cosine and B-spline representations are linear in their coefficients, $f = \mathsf{B}\boldsymbol{c}$, whereas the network and MixFunn are not. This difference matters: for a given trajectory, the residual of Eq. (2) is linear in $\boldsymbol{c}$, so the

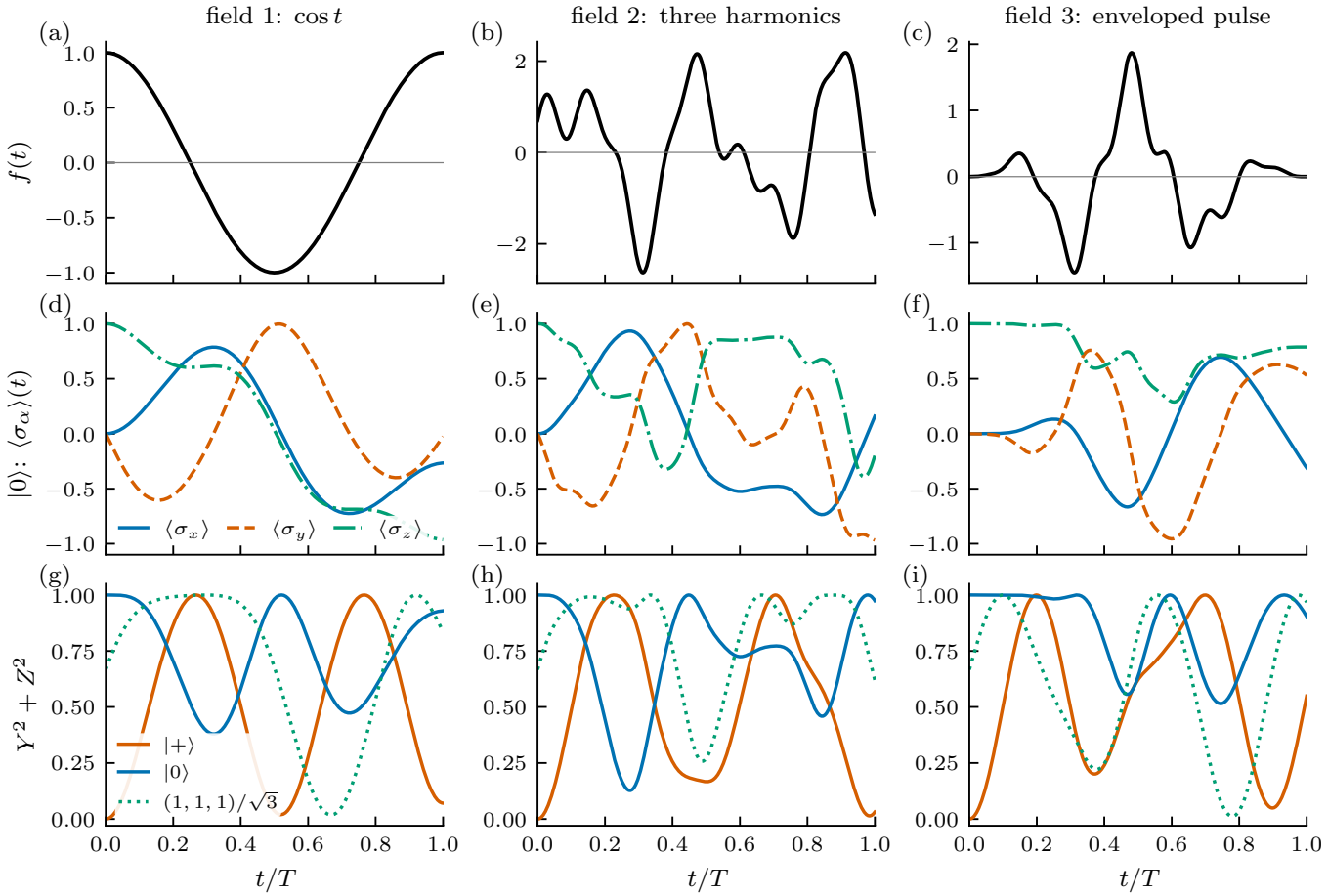


FIG. 1. The inverse problem. (a)–(c) Test fields. (d)–(f) Bloch-vector components for the initial state $|0\rangle$, which are the data (without noise). (g)–(i) Sensitivity of the data to the field, $Y^2 + Z^2 = 1 - X^2$, for three initial states: with $|+\rangle$, an eigenstate of the control operator σ_x, it vanishes at $t = 0$ and repeatedly afterwards, while with $|0\rangle$ it stays above 0.13.

optimal coefficients follow from a linear least-squares problem. This enables the classical estimators of Sec. IV C, a least-squares initialization of the PINN, and variable projection [29].

IV. ESTIMATION METHODS

A. Physics-informed neural network

A network $\mathcal{N}_{\boldsymbol{\theta}}(s)$ with sine activations and normalized input $s = 2t/T - 1$ represents the state through

$$\boldsymbol{r}_{\boldsymbol{\theta}}(t) = \boldsymbol{r}_0 + \frac{t}{T}\,\mathcal{N}_{\boldsymbol{\theta}}(s), \tag{4}$$

which satisfies the initial condition exactly [6, 30, 31]. When the values $a = f(0)$ and $b = f(T)$ are known from the design of the pulse (field 3 starts and ends at zero), they are also imposed exactly, $f(t) = g(t) - (1-\tau)g(0) - \tau g(T) + (1-\tau)a + \tau b$ with $\tau = t/T$ and g the unconstrained representation; this transformation is linear in g and therefore preserves the linearity of the basis representations. The loss is

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}|}\sum_{t\in\mathcal{C}} \left\|\dot{\boldsymbol{r}}_{\boldsymbol{\theta}} - \boldsymbol{\Omega}_{\boldsymbol{\theta}} \times \boldsymbol{r}_{\boldsymbol{\theta}}\right\|^2 + \frac{w}{N}\sum_{i=1}^{N} \left\|\boldsymbol{r}_{\boldsymbol{\theta}}(t_i) - \boldsymbol{r}_i\right\|^2 + \frac{\lambda_f}{T}\int_0^T \left(f''_{\boldsymbol{\theta}}\right)^2 dt, \tag{5}$$

where the collocation set $\mathcal{C}$ contains the N data times and 200 uniform random times redrawn at every Adam epoch [6, 32], $w = 1$, and the smoothness weight λ_f is fixed a priori by the noise level ($\lambda_f = 0$ without noise). Time derivatives are propagated through the layers in forward mode, $\dot{\boldsymbol{h}}_{l+1} = \sigma'(\mathsf{W}_l \boldsymbol{h}_l + \boldsymbol{b}_l) \odot \mathsf{W}_l \dot{\boldsymbol{h}}_l$, which is exact and cheaper than nested automatic differentiation. The loss is exactly the squared norm of a residual vector, $\mathcal{L} = \|\boldsymbol{\rho}(\boldsymbol{\theta})\|^2$.

B. Optimization

All PINNs start with 3000 Adam steps [33]. For comparison, we also reproduce the common practice of training with Adam alone (smaller network, raw time input, residual at the data points only, constant learning rate, single precision). After Adam, a second-order stage is applied: L-BFGS [34], a dense BFGS update with or without self-scaling [12, 13, 35], or the LM method [36, 37],

$$\delta\boldsymbol{\theta} = -\left(\mathsf{J}^\top\mathsf{J} + \mu\mathsf{I}\right)^{-1}\mathsf{J}^\top\boldsymbol{\rho} = -\mathsf{J}^\top\left(\mathsf{J}\mathsf{J}^\top + \mu\mathsf{I}\right)^{-1}\boldsymbol{\rho}, \quad (6)$$

with $\mathsf{J} = \partial\boldsymbol{\rho}/\partial\boldsymbol{\theta}$ and the damping μ updated by Nielsen's rule [38]. LM is a damped Gauss–Newton method; in the PINN context the Gauss–Newton direction coincides with the energy natural gradient [14, 15], and the second form of Eq. (6), used when parameters outnumber residuals, is its residual-space version [16]. The Jacobian is assembled from per-point Jacobians, at a cost proportional to the number of points times the number of parameters. We also tested causal weights [39], gradient-norm loss balancing [40] and variable projection. [**TODO:** ablation results (profile `ablacao`); SM]

C. Classical estimators

Direct inversion evaluates Eq. (3) with derivatives of an interpolating cubic spline (or of a smoothing spline with the smoothing parameter chosen by generalized cross-validation for noisy data [17]). *Linear least squares* (LSQ) minimizes the residual of the second and third Eqs. (2) over the coefficients of a basis, with the same spline derivatives and a second-difference penalty $\lambda\|\mathsf{D}_2\boldsymbol{c}\|^2$ for noisy data [17, 27]. *Shooting* integrates Eq. (2) for a B-spline field (fourth-order Runge–Kutta with forward sensitivities [41]) and fits the coefficients to the data with LM, starting from the LSQ solution; it is the estimation counterpart of gradient-based pulse design [42] and, unlike the PINN, satisfies the dynamics exactly.

D. Benchmark protocol

All methods use the same data for each field, noise level and seed; the true field never enters a loss or the choice of a hyper-parameter; regularization weights depend only on the noise level; every configuration is run with five seeds, which set the network initialization, the noise realization and the random collocation points. We report the relative error of the field on the data grid, $\epsilon_f = \|f_{\mathrm{pred}} - f\|_2/\|f\|_2$, and the RMS error of the trajectory re-simulated with f_{pred}, ϵ_r. Each PINN run is limited to 420 s on one CPU thread. All settings are listed in the SM.

V. RESULTS

Table I summarizes the median field error of the main methods. Figures 2–4 show, for each field, the recovered field, its pointwise error, and both metrics for every method; per-seed results and run times are given in the SM.

A. Field 1: a single harmonic

For $f = \cos t$ all linear representations contain the exact field, and every estimator that optimizes well reaches errors of 10^{-5}–10^{-4} (Fig. 2): 4×10^{-6} for LSQ with the Fourier basis, 1–4×10^{-5} for the PINN with a Levenberg–Marquardt stage. MixFunn, whose dictionary contains cos, reaches 3×10^{-4}. The Adam-only PINN stops at 5×10^{-3}: even in the easiest case it is limited by the optimizer. Direct inversion is limited to 10^{-3} by the spline derivatives near the edges.

B. Field 2: a non-periodic multi-harmonic field

Field 2 does not satisfy $f(0) = f(T)$, and the periodic Fourier representation fails regardless of the estimator: $\epsilon_f = 0.16$ for LSQ and 0.18 for the PINN, with the error concentrated at the edges (Fig. 3). The cosine series reduces the error by an order of magnitude (1.6×10^{-2} for LSQ) and cubic B-splines with $n = 80$ functions by two (1.2×10^{-3}). With an LM stage, the PINN with a network field reaches 3.1×10^{-3}, comparable to shooting (3.7×10^{-3}) and direct inversion (3.0×10^{-3}). Without it, the PINN error is $0.22^\dagger$. MixFunn did not converge within the time limit (0.13).

C. Field 3: a smooth pulse, with and without noise

For field 3 the Adam-only PINN converges to an oversmoothed field with $\epsilon_f = 0.26^\dagger$ [Fig. 4(a)], although the re-simulated trajectory is already close to the data ($\epsilon_r \sim 10^{-2}$): a small trajectory error does not imply an accurate field, in line with the observation that PINN solutions become accurate only at very small loss values [11]. With an LM stage the error drops to (1.8–2.0) $\times10^{-3}$ for the network, cosine and B-spline representations, and to 5.4×10^{-3} for the Fourier series, which is limited by its basis (the same value as Fourier LSQ). The classical estimators with a rich basis are about ten times more accurate: 2.5×10^{-4} for LSQ with B-splines or cosines and 1.5×10^{-4} for shooting. Panel (c) shows that both metrics rank the methods in the same order once the optimizer has converged.

With noise ($\sigma = 0.01$) all estimators that converge lie between 4×10^{-2} and 7×10^{-2} [Fig. 4(d)–(f)]; the error is dominated by the noise, mostly in the high-frequency components to which the data are least sensi-

TABLE I. Median relative field error ϵ_f over five seeds. LSQ: linear least squares. †Not yet available in the current protocol; value from an earlier protocol with a fixed collocation grid, 40 B-splines and no edge constraint (see SM). [**TODO:** complete benchmark (180 of 440 runs missing)]

	method	field 1	field 2	field 3	field 3, $\sigma = 0.01$
classical	direct inversion	1.3×10^{-3}	3.0×10^{-3}	1.6×10^{-3}	6.9×10^{-2}
	LSQ, Fourier (M=10)	4.1×10^{-6}	0.16	4.9×10^{-3}	3.9×10^{-2}
	LSQ, cosine (M=40)	5.6×10^{-6}	1.6×10^{-2}	2.5×10^{-4}	6.5×10^{-2}
	LSQ, B-spline (n=80)	8.6×10^{-6}	1.2×10^{-3}	2.5×10^{-4}	6.7×10^{-2}
	shooting, B-spline (n=80)	2.4×10^{-5}	3.7×10^{-3}	1.5×10^{-4}	5.7×10^{-2}
PINN	NN field, Adam only	$5.5 \times 10^{-3\dagger}$	$0.22^{\dagger}$	$0.26^{\dagger}$	$0.26^{\dagger}$
	NN field, Adam + L-BFGS	$7.3 \times 10^{-3\dagger}$	$0.14^{\dagger}$	$3.3 \times 10^{-2\dagger}$	$6.0 \times 10^{-2\dagger}$
	NN field, Adam + LM	3.9×10^{-5}	3.1×10^{-3}	2.0×10^{-3}	$5.5 \times 10^{-2\dagger}$
	Fourier (M=10), Adam + LM	7.8×10^{-6}	0.18	5.4×10^{-3}	$4.8 \times 10^{-2\dagger}$
	cosine (M=40), Adam + LM	1.3×10^{-5}	2.4×10^{-2}	1.9×10^{-3}	–
	B-spline (n=80), Adam + LM	3.6×10^{-5}	5.1×10^{-3}	1.8×10^{-3}	$5.5 \times 10^{-2\dagger}$
	MixFunn, Adam only	5.3×10^{-2}	0.89	0.65	0.58
	MixFunn, Adam + LM	2.7×10^{-4}	0.13	0.25	0.26

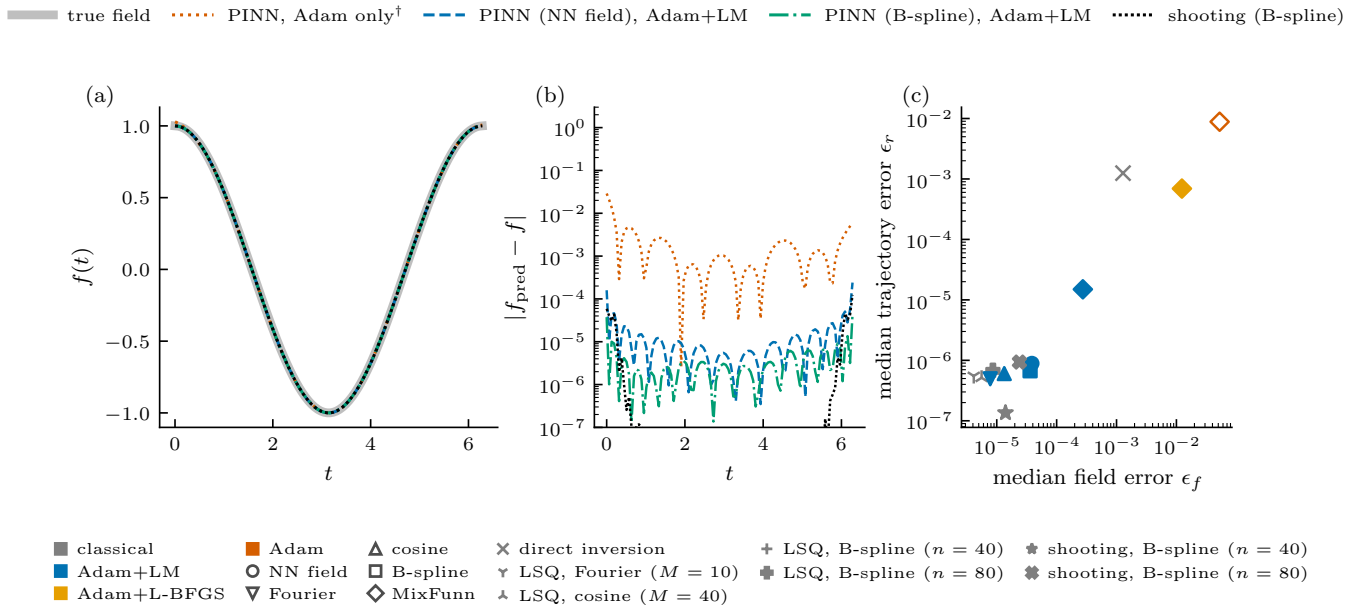


FIG. 2. Field 1, $f = \cos t$, without noise. (a) True field and estimates (seed 0); (b) pointwise error; (c) median field error ϵ_f versus median trajectory error ϵ_r for every method (colour: optimizer; marker: field representation). †Earlier protocol (see Table I).

tive. The Fourier basis, with only 21 coefficients, acts as a strong regularizer and gives the lowest error (3.9×10^{-2}). [**TODO:** PINN with LM stage and noise in the current protocol; regularization chosen by validation.]

D. The optimizer is the bottleneck

Figure 5 isolates the effect of the second stage for field 3, with the network and the B-spline representations and the same 3000 Adam steps. L-BFGS lowers the median error from 0.26 to 3×10^{-2}, dense BFGS to 10^{-2}, and LM to 10^{-3} (network field). Our self-scaled BFGS variant did not reproduce the gains reported in Refs. [12, 13] ($\simeq 0.1$; see SM), and causal weights [39] did not improve LM. These results agree with the ill-conditioning analysis of Ref. [11]: for this problem, the choice of optimizer changes the error by two orders of magnitude, much more than the choice of representation.

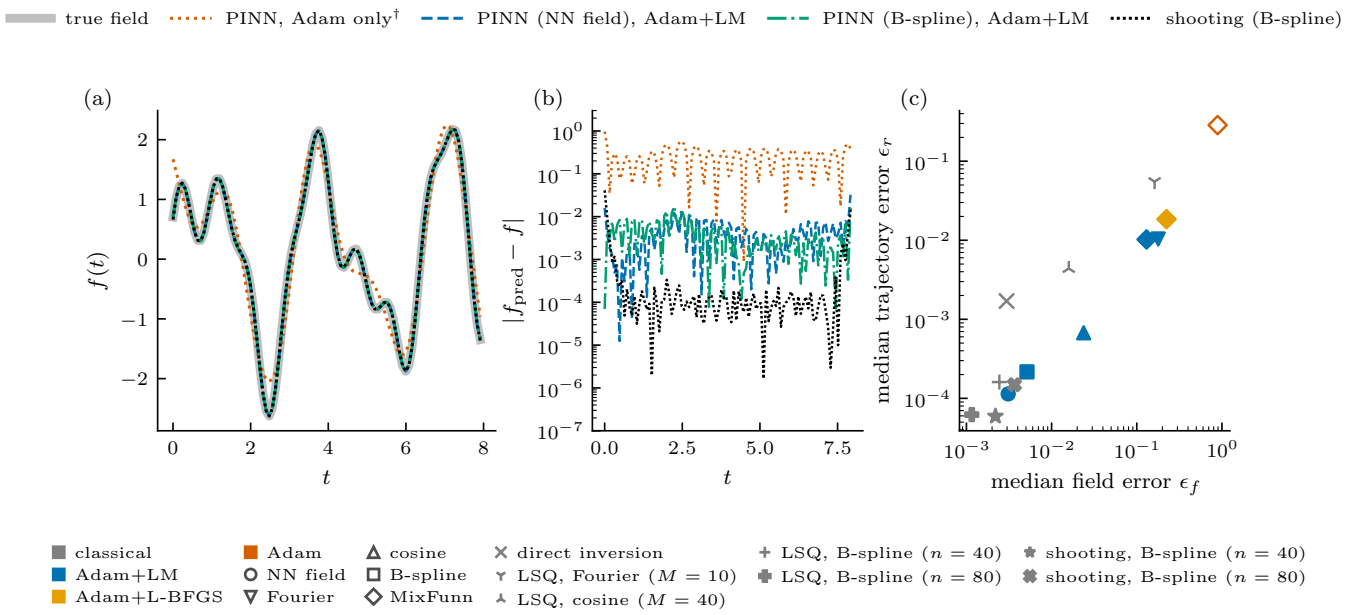


FIG. 3. Field 2, without noise; panels as in Fig. 2. The periodic Fourier basis produces Gibbs oscillations at the edges.

VI. OUTLOOK: HARDER SETTINGS

The results above concern the most favourable case: a single qubit whose full Bloch vector is measured. There, classical estimators are hard to beat. The value of a PINN lies where these estimators do not apply directly; we identify two such settings and the obstacles they bring.

A. Two-qubit Hamiltonian

For two qubits with a time-dependent exchange coupling,

$$H(t) = \omega_1\,\sigma_z^{(1)} + \omega_2\,\sigma_z^{(2)} + f(t)\big(\sigma_x^{(1)}\sigma_x^{(2)} + \sigma_y^{(1)}\sigma_y^{(2)}\big), \quad (7)$$

the data are the fifteen expectation values $\langle\sigma_\mu^{(1)}\sigma_\nu^{(2)}\rangle$, which obey the Heisenberg equations used in the PINNverse framework [7, 8]; amplitude damping and dephasing can be included through Lindblad dissipators [43, 44]. Two features make this problem qualitatively different. (i) The exchange term conserves the number of excitations, and $(\sigma_x\sigma_x + \sigma_y\sigma_y)|00\rangle = 0$: the initial states $|00\rangle$ and $|11\rangle$ are completely blind to the field, the two-qubit analogue of $|+\rangle$ in Sec. II B. We verified numerically that the fifteen expectation values are identical for f and $f/2$ when starting from $|00\rangle$, whereas they differ by order one from $|01\rangle$ or $|{+}{+}\rangle$. (ii) For equal frequencies, $\omega_1 = \omega_2$, the drift commutes with the coupling and $U(t) = e^{-iH_0t}e^{-i\Phi(t)H_1}$ with $\Phi(t) = \int_0^t f$; the data depend on f only through its integral (verified to 10^{-7}). The field must then be recovered as the derivative of an identified phase, which changes the conditioning of the problem with respect to the single-qubit case. These two-qubit models, with and without dissipation, are implemented in the original notebooks of our code repository and will be included in the protocol.

B. Partial information

If only $\langle\sigma_z\rangle$ is measured, Eq. (3) and the linear least-squares estimators are no longer available, because X and Y are unknown. A PINN treats them as latent variables constrained by Eq. (2), and shooting remains applicable. The symmetry $\sigma_z H(f)\sigma_z = H(-f)$, together with $\sigma_z|0\rangle = |0\rangle$, implies that f and $-f$ produce exactly the same $\langle\sigma_z\rangle(t)$ (verified numerically), so the sign of the field is not identifiable from $|0\rangle$. A second symmetry of Eq. (2), $(X, Y, Z, f) \to (X, Y, -Z, -f)$, leaves invariant every initial state with $Z = 0$; from $|+\rangle$, therefore, $\langle\sigma_x\rangle$ and $\langle\sigma_y\rangle$ together still cannot fix the sign of f. A generic initial state, e.g. $(1, 1, 1)/\sqrt{3}$, breaks both symmetries ($\max_t |\Delta Z| = 1.5$ between f and $-f$), as does measuring two observables. Several initial states sharing the same field remove the blind intervals of each one; with the same measurement budget, three initial states reduced the LSQ error from 0.15 to 0.07 in a preliminary study (Table S3). The spectral resolution of each method—the error as a function of the frequency of the field, whose limits are set by the $1/\omega$ sensitivity and by the basis (Figs. S1 and S2)—will be quantified with single-tone pulses. [**TODO:** run profile `espectral` and the partial-data extension]

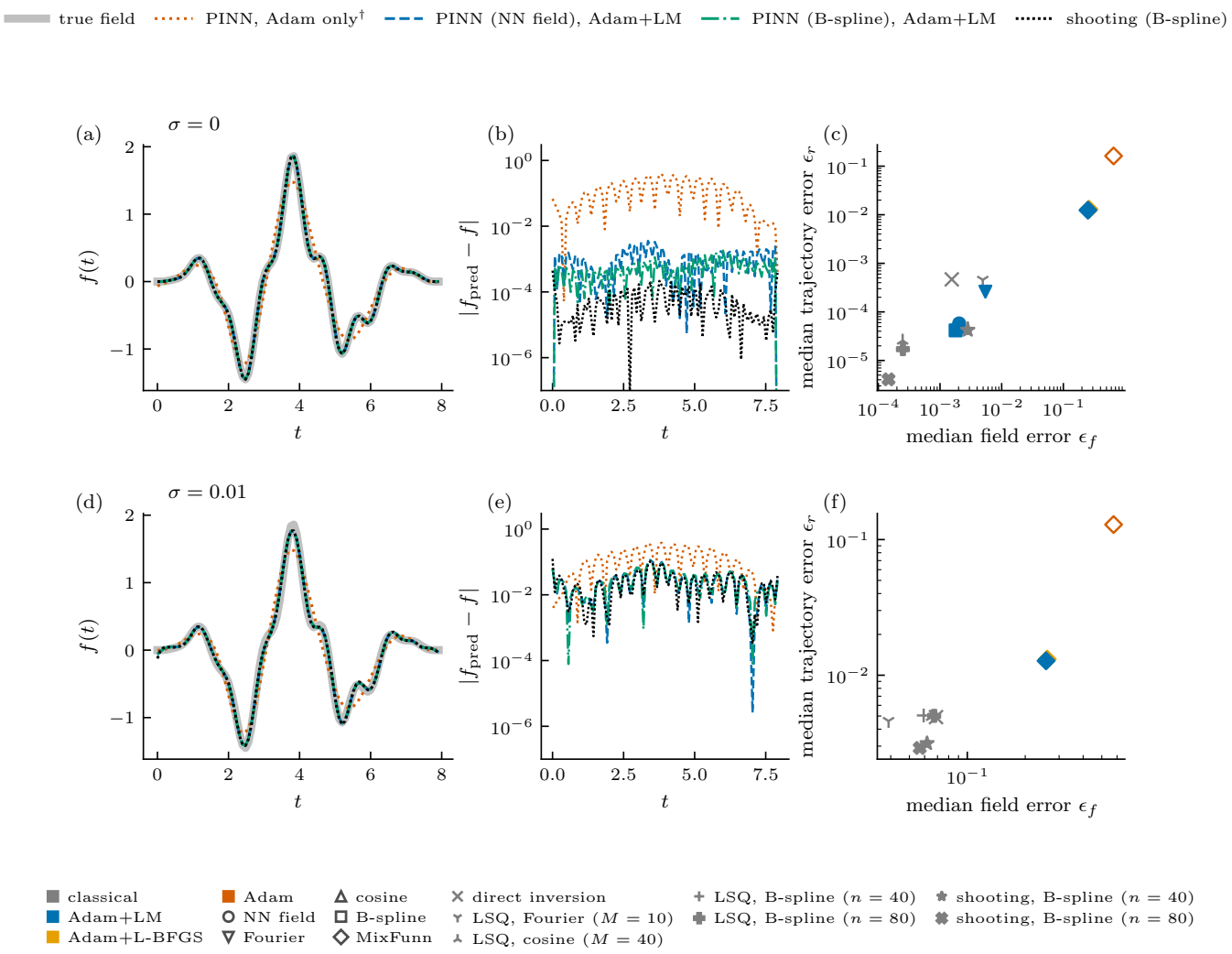


FIG. 4. Field 3, (a)–(c) without noise and (d)–(f) with noise $\sigma = 0.01$; panels as in Fig. 2.

VII. CONCLUSIONS

We compared PINNs with classical estimators for recovering a time-dependent control field from qubit expectation values under a single protocol. The initial state must not be aligned with the control axis; $|0\rangle$ makes the field observable at all times. Trained with Adam alone, PINNs converge to an over-smoothed field whatever the representation of the field, while the trajectory already fits the data. A Gauss–Newton (Levenberg–Marquardt) stage removes this limitation and lowers the error by two orders of magnitude, after which the representation matters: periodic Fourier series fail for non-periodic fields, and cubic B-splines, cosine series and network fields perform similarly. With complete, noiseless data, classical least squares and shooting with a rich basis remain more accurate and much cheaper; with noise, all good estimators are limited by the data. The natural place for physics-informed estimators is therefore the setting where classical estimators fail—partial measurements, several qubits and open-system dynamics—provided that the blind states and symmetries identified here are avoided and the training uses a second-order optimizer. [**TODO:** revise when the benchmark is complete]

ACKNOWLEDGMENTS

[**TODO:** funding agencies (FAPESP, CNPq, CAPES) and grant numbers.]

DATA AVAILABILITY

The code, the data of every run and the scripts that produce all figures and tables are available at `https://github.com/GubioGL/Pinn-inverse-for-H-time`.

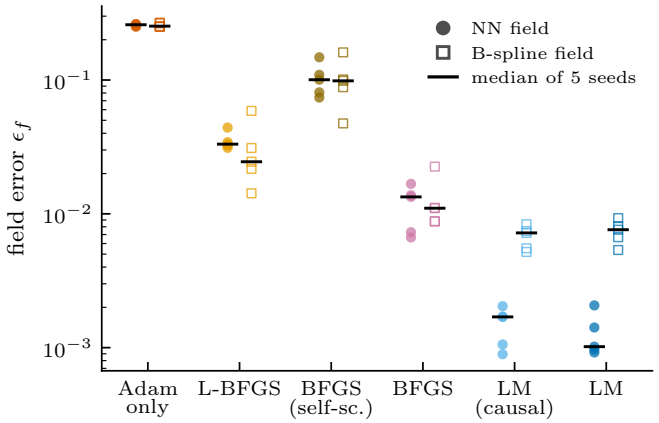


FIG. 5. Field error for field 3 (no noise) after 3000 Adam steps followed by each second-stage optimizer; points are seeds, bars are medians. Data from the earlier protocol (fixed collocation grid, 40 B-splines). [**TODO:** update with profile `otimizadores`]

[1] S. Gustavsson, O. Zwier, J. Bylander, F. Yan, F. Yoshihara, Y. Nakamura, T. P. Orlando, and W. D. Oliver, Phys. Rev. Lett. **110**, 040502 (2013).

[2] J. Zhang and M. Sarovar, Phys. Rev. Lett. **113**, 080401 (2014).

[3] N. Wiebe, C. Granade, C. Ferrie, and D. G. Cory, Phys. Rev. Lett. **112**, 190501 (2014).

[4] M. Raissi, P. Perdikaris, and G. E. Karniadakis, J. Comput. Phys. **378**, 686 (2019).

[5] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, Nat. Rev. Phys. **3**, 422 (2021).

[6] A. Norambuena, M. Mattheakis, F. J. González, and R. Coto, Phys. Rev. Lett. **132**, 010801 (2024).

[7] L. K. Castelano, I. Cunha, F. S. Luiz, R. de Jesus Napolitano, M. V. S. Prado, and F. F. Fanchini, Phys. Rev. A **110**, 032607 (2024).

[8] G. G. de Lima, I. Cunha, and L. K. Castelano, Inverse physics-informed neural networks procedure for detecting noise in open quantum systems (2025), arXiv:2507.12552 [quant-ph].

[9] J. Liu and X. Wang, Phys. Rev. Res. **7**, 043137 (2025), arXiv:2506.10379.

[10] A. S. Krishnapriyan, A. Gholami, S. Zhe, R. M. Kirby, and M. W. Mahoney, in *Advances in Neural Information Processing Systems*, Vol. 34 (2021).

[11] P. Rathore, W. Lei, Z. Frangella, L. Lu, and M. Udell, in *Proceedings of the 41st International Conference on Machine Learning*, PMLR, Vol. 235 (2024) pp. 42159–42191.

[12] J. F. Urbán, P. Stefanou, and J. A. Pons, J. Comput. Phys. (2025), arXiv:2405.04230.

[13] E. Kiyani, K. Shukla, J. F. Urbán, J. Darbon, and G. E. Karniadakis, Which optimizer works best for physics-informed neural networks and Kolmogorov-Arnold networks? (2025), arXiv:2501.16371.

[14] J. Müller and M. Zeinhofer, in *Proceedings of the 40th International Conference on Machine Learning*, PMLR, Vol. 202 (2023).

[15] A. Jnini, F. Vella, and M. Zeinhofer, Gauss-newton natural gradient descent for physics-informed computational fluid dynamics (2024), arXiv:2402.10680.

[16] A. Jnini and F. Vella, Dual natural gradient descent for scalable training of physics-informed neural networks (2025), arXiv:2505.21404.

[17] P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms* (SIAM, Philadelphia, 2010).

[18] C. de Boor, *A Practical Guide to Splines*, revised ed. (Springer, New York, 2001).

[19] T. de Souza Farias, G. G. de Lima, J. Maziero, and C. J. Villas-Boas, MixFunn: A neural network for differential equations with improved generalization and interpretability (2025), arXiv:2503.22528.

[20] F. Bloch, Phys. Rev. **70**, 460 (1946).

[21] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. (Cambridge University Press, Cambridge, 2010).

[22] G. Cybenko, Math. Control Signals Syst. **2**, 303 (1989).

[23] K. Hornik, M. Stinchcombe, and H. White, Neural Netw. **2**, 359 (1989).

[24] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville, in *Proceedings of the 36th International Conference on Machine Learning*, PMLR, Vol. 97 (2019) pp. 5301–5310.

[25] M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng, in *Advances in Neural Information Processing Systems*, Vol. 33 (2020).

[26] J. P. Boyd, *Chebyshev and Fourier Spectral Methods*, 2nd ed. (Dover, New York, 2001).

[27] P. H. C. Eilers and B. D. Marx, Stat. Sci. **11**, 89 (1996).

[28] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, KAN: Kolmogorov-Arnold networks (2024), arXiv:2404.19756.

[29] G. H. Golub and V. Pereyra, SIAM J. Numer. Anal. **10**, 413 (1973).

[30] I. E. Lagaris, A. Likas, and D. I. Fotiadis, IEEE Trans. Neural Netw. **9**, 987 (1998).

[31] L. Lu, R. Pestourie, W. Yao, Z. Wang, F. Verdugo, and S. G. Johnson, SIAM J. Sci. Comput. **43**, B1105 (2021).

[32] C. Wu, M. Zhu, Q. Tan, Y. Kartha, and L. Lu, Comput. Methods Appl. Mech. Eng. **403**, 115671 (2023).

[33] D. P. Kingma and J. Ba, Adam: A method for stochastic optimization (2015), iCLR 2015, arXiv:1412.6980.

[34] D. C. Liu and J. Nocedal, Math. Program. **45**, 503 (1989).

[35] J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. (Springer, New York, 2006).

[36] K. Levenberg, Q. Appl. Math. **2**, 164 (1944).

[37] D. W. Marquardt, J. Soc. Ind. Appl. Math. **11**, 431 (1963).

[38] H. B. Nielsen, *Damping Parameter in Marquardt's Method*, Tech. Rep. IMM-REP-1999-05 (Informatics and Mathematical Modelling, Technical University of Denmark, 1999).

[39] S. Wang, S. Sankaran, and P. Perdikaris, Respecting causality is all you need for training physics-informed neural networks (2022), arXiv:2203.07404.

[40] S. Wang, Y. Teng, and P. Perdikaris, SIAM J. Sci. Comput. **43**, A3055 (2021).

[41] E. Hairer, S. P. Nørsett, and G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed. (Springer, Berlin, 1993).

[42] N. Khaneja, T. Reiss, C. Kehlet, T. Schulte-Herbrüggen, and S. J. Glaser, J. Magn. Reson. **172**, 296 (2005).

[43] G. Lindblad, Commun. Math. Phys. **48**, 119 (1976).

[44] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, J. Math. Phys. **17**, 821 (1976).